Topics in Algebra (FA24)

Jump to Today

Course Meetings:

- **Lectures:** Tuesday/Thursday 2:25pm - 4:15pm (2A)
- **X-period:** Wednesday 5:30pm - 6:20pm
- **X Period Usage:** The X-period will be used and scheduled as needed to supplement, extend, or make-up our normal course lectures at my discretion. When scheduled students are expected to attended the X periods.
- **Room:** Kemeny 006

Instructor Information:

- **Instructor:** Juliette Bruce
- **Instructor office:** Kemeny 342
- **E-mail:** juliette.bruce@dartmouth.edu (mailto:juliette.bruce@dartmouth.edu)
- **Office hours:** Wednesdays 1:30pm-2:30pm (Office) and by appointment

Math With Friends:

Our undergraduate TA (Lisa Samoylov) will hold a Math With Friends drop-in study group (no appointments needed) on Sundays 4:30-7:30pm and Wednesdays 7:30-9:30pm in Kemeny 121.

Other Information:

- **Prerequisites:** Math 22 or 24. If you are unsure about your preparation, please talk to me!
- **Required Text:** *A Book of Abstract Algebra - 2nd edition* by Charles C. Pinter.

Grading:

- **Grade Breakdown:** Grades will be based on:
  - Weekly Homework (40%),
  - In-Classes Quizzes (10%),
  - In-Class Midterm Exam (25%), and
  - Final Project (25%).
- **Dropped Scores:** Your lowest homework score and your lowest quiz score will be dropped.

You are currently logged into Student View

Resetting the test stud course as a brand new

Midterm:

Our midterm is in-class on a date to be determined.

Final Project:

The final project is due on November 19, 2024 both in class and on Canvas.

Homework:

Homework is due in-person at the start of class on the due date, free to submit handwritten or work typeset in LaTeX (**Overleaf (https://www.overleaf.com/learn/latex/Learn_LaTeX_in_30_minutes)** is a great resource and tool for the latter). You are more than welcome to work with others on the homework problems. However, the solutions should be written up *on your own and reflect your understanding*.

Course Schedule:

The course schedule below will be updated as we proceed through the semester:

Week	Topics	Pinter References
Week 1	Why Abstract Algebra?, Sets & Proofs, Operations & Groups	Chapters 1, 2, Appendix A
Week 2	Groups, Group Properties, Subgroups	Chapters 3, 4, 5
Week 3	Functions, Permutation Groups, Cycles Decomposition	Chapters 6, 7, 8
Week 4	Examples of Groups, Isomorphism	Chapters 8, 9, 10, 11
Week 5	Equivalence Relations	Chapters 12

Week 6	Cosets, Homomorphisms, Quotients	Chapters 13, 14, 15, 16
Week 7	Group Actions, Rings, Ideals	Chapter 18, Appendix B
Week 8	Quotient Rings, Domains, Primes	Chapters 19. 20, 21, 22
Week 9	Polynomial Rings	Chapters 24, 25, 26

Course Expectations:

I believe in the axioms laid out by SFSU Professor* ***Federico Ardila*** ***(http://fardila.com/)*** *, and I will use them to guide my instruction of this course.*

1. Mathematical potential is distributed equally among different groups, irrespective of geographic, demographic, and economic boundaries.

2. Everyone can have joyful, meaningful, and empowering mathematical experiences.

3. Mathematics is a powerful, malleable tool that can be shaped and used differently by various communities to serve their needs.

4. Every student deserves to be treated with dignity and respect.

*Mathematically, an axiom is a statement accepted as truth, on which all other work is built.

Using these as guiding principles, I expect students in Math 31 to collaborate with me in developing and maintaining an inclusive learning environment where diversity and individual differences are understood, respected, and recognized as a source of strength. Racism, discrimination, harassment, and bullying will not be tolerated. I expect all participants in this course (students and faculty alike) to treat each other with courtesy and respect.

Mathematics requires active participation. Before each class period, please read the assigned sections. Arrive to class ready to share what you have learned and also what remains confusing. Class meetings will involve some interactive lecture and other activities in a variety of formats; you will get the most out of each class day if you arrive ready to engage. In all settings, collaborate thoughtfully and ask questions respectfully: everyone should be able to freely participate.

Course Catalogue Description:

This course will provide an introduction to fundamental algebraic structures, and may include significant applications. The majority of the course will consist of an introduction to the basic algebraic structures of groups and rings. Additional work will consist either of the development of further algebraic structures or applications of the previously developed theory to areas such as coding theory or crystallography.

As a result of the variable syllabus, this course may not serve as an adequate prerequisite for **MATH 81** (http://dartmouth.smartcatalogiq.com/en/current/orc/Departments-Programs-Undergraduate/Mathematics/MATH-Mathematics-Undergraduate/MATH-81) . Students who contemplate taking **MATH 81** (http://dartmouth.smartcatalogiq.com/en/current/orc/Departments-Programs-Undergraduate/Mathematics/MATH-Mathematics-Undergraduate/MATH-81) should consider taking **MATH 71** (http://dartmouth.smartcatalogiq.com/en/current/orc/Departments-Programs-Undergraduate/Mathematics/MATH-Mathematics-Undergraduate/MATH-71) instead of this course.

Learning Outcomes:

By the end of this course, you should be able to:

1. Understand of the basic structures of algebra: define terms, explain their significance, and apply them in context;
2. Solve mathematical problems: utilize abstraction and think creatively;
3. Write clear mathematical proofs: recognize and construct mathematically rigorous arguments.

Academic Honor Principal:

Cooperation on weekly homework is permitted (and encouraged), but you should write up the solution on your own. In other words, share your thoughts (say on a blackboard) to get started, but write up the solution afterwards using only your understanding. Please acknowledge any cooperative work at the end of each assignment.

Plagiarism, collusion, or other violations of the **Academic Honor Principle** (https://students.dartmouth.edu/community-standards/policy/academic-honor-principle) will be referred to the The Committee on Standards. If you have any questions as to whether some action would be acceptable under the Academic Honor Principle, please speak to me beforehand. For more information about standards of conduct, please see **these resources** (https://students.dartmouth.edu/community-standards/) .

Mental Health:

The academic environment at Dartmouth is challenging, our terms are intensive, and classes are not the only demanding part of your life. There are a number of resources available to you on campus to support your wellness:

- Undergraduate Dean (http://www.dartmouth.edu/~upperde/ (http://www.dartmouth.edu/~upperde/));
- Counseling and Human Development (http://www.dartmouth.edu/~chd/ (http://www.dartmouth.edu/~chd/)); and the
- Student Wellness Center (http://www.dartmouth.edu/~healthed/ (http://www.dartmouth.edu/~healthed/)).

I encourage you to use these resources to take care of yourself throughout the term. Please come speak to me if you experience any difficulties, or would like help accessing any of these recourses.

Student Accessibility Services:

Students requesting disability-related accommodations and services for this course are encouraged to schedule a phone/video meeting with me as early in the term as possible. This conversation will help to establish what supports are built into my online course. In order for accommodations to be authorized, students are required to consult with Student Accessibility Services (https://students.dartmouth.edu/student-accessibility/) (SAS; student.accessibility.services@dartmouth.edu (mailto:student.accessibility.services@dartmouth.edu) ; 603-646-9900) and to email me their SAS accommodation form. We will then work together with SAS if accommodations need to be modified based on the online learning environment. If students have questions about whether they are eligible for accommodations, they should contact the SAS office. All inquiries and discussions will remain confidential.

Title IX:

At Dartmouth, we value integrity, responsibility, and respect for the rights and interests of others, all central to our Principles of Community. We are dedicated to establishing and maintaining a safe and

inclusive campus where all have equal access to the educational and employment opportunities Dartmouth offers. We strive to promote an environment of sexual respect, safety, and well-being. In its policies and standards, Dartmouth demonstrates unequivocally that sexual assault, gender-based harassment, domestic violence, dating violence, and stalking are not tolerated in our community.

The Sexual Respect Website (**https://sexual-respect.dartmouth.edu (https://sexual-respect.dartmouth.edu/)**) at Dartmouth provides a wealth of information on your rights with regard to sexual respect and resources that are available to all in our community.

Please note that, as a faculty member, I am obligated to share disclosures regarding conduct under Title IX with Dartmouth's Title IX Coordinator. Confidential resources are also available, and include licensed medical or counseling professionals (e.g., a licensed psychologist), staff members of organizations recognized as rape crisis centers under state law (such as WISE), and ordained clergy (see **https://dartgo.org/titleix_resources (https://dartgo.org/titleix_resources)**).

Should you have any questions, please feel free to contact Dartmouth's Title IX Coordinator or the Deputy Title IX Coordinator for the Guarini School. Their contact information can be found on the sexual respect website at: **https://sexual-respect.dartmouth.edu (https://sexual-respect.dartmouth.edu/)** .

Religous Observances:

Some students may wish to take part in religious observances that occur during this academic term. If you have a religious observance that conflicts with your participation in the course, please meet with me as soon as possible, or before the end of the second week of the term (at the latest) to discuss appropriate adjustments. Dartmouth has a deep commitment to support students' religious observances and diverse faith practices.

Additional Support:

Cooperation on weekly homework is permitted (and encouraged), but you should write up the solution on your own. In other words, share your thoughts (say on a blackboard) to get started, but write up the solution afterwards using only your understanding. Please acknowledge any cooperative work at the end of each assignment.

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Course Summary:

Date	Details	Due
Tue Sep 24, 2024	Quiz 1 (https://canvas.dartmouth.edu/courses/67870/assignments/485151)	due by 11:59pm
Tue Oct 1, 2024	Homework 1 (https://canvas.dartmouth.edu/courses/67870/assignments/485149)	due by 2:25pm
	Quiz 2 (https://canvas.dartmouth.edu/courses/67870/assignments/486150)	due by 11:59pm
Tue Oct 15, 2024	Homework 2 (https://canvas.dartmouth.edu/courses/67870/assignments/486581)	due by 11:59pm
Thu Oct 17, 2024	Quiz 3 (https://canvas.dartmouth.edu/courses/67870/assignments/487572)	due by 11:59pm
Thu Oct 24, 2024	Homework 3 (https://canvas.dartmouth.edu/courses/67870/assignments/487546)	due by 11:59pm
Tue Oct 29, 2024	Midterm (https://canvas.dartmouth.edu/courses/67870/assignments/487554)	due by 11:59pm
Tue Nov 5, 2024	Quiz 4 (https://canvas.dartmouth.edu/courses/67870/assignments/489811)	due by 11:59pm
Tue Nov 12, 2024	Quiz 5 (https://canvas.dartmouth.edu/courses/67870/assignments/490645)	due by 11:59pm
	Homework 4	

	(https://canvas.dartmouth.edu/courses/67870/assignments/493157)	due by 11:59pm
Thu Nov 14, 2024		
	Quiz 6 (https://canvas.dartmouth.edu/courses/67870/assignments/491013)	due by 11:59pm
Tue Nov 19, 2024	Final Project (https://canvas.dartmouth.edu/courses/67870/assignments/485698)	due by 11:59pm